

Chapter 7

Queueing Models

7.1 Introduction

In this chapter we continue studying operations in the time dimension. For example, in a production activity or a warehouse operation, the order that tasks are performed affects the final result. For example, in a production line, some stations might process jobs faster than others, and the throughput will be affected by how the production line is configured. In a warehouse operation, when many trucks arrive at the same time, the number of loading stations and workers will affect the effective throughput of the operation. Knowing how to assign and configure manufacturing systems to be faster and more resilient is a key strategic tool to develop supply chain operations.

Mathematically, we leverage our study of networks and probabilistic theory to describe common problems in queueing systems.

7.2 A network view of a queueing system

Queueing systems are defined as a probabilistic system where jobs, customers or tasks *evolve* inside a well limited system: for example, a production line queue, cars passing through a bridge or a flower plantation.

The primitive unit in all these systems is the number of jobs (or customers, tasks, etc) in the system. At any time t the number of jobs $N = 0, 1, 2, \dots$ is a unique value, this feature makes us think that in time-homogeneous systems (the arrivals, processing and departures are time-independent), we can think of a probabilistic distribution of observing the number in system as a proportion of time:

$$\pi_i := \lim_{t \rightarrow \infty} \int_0^t \frac{\mathbf{1}\{X_s = i\}}{t} ds \quad (7.1)$$

Where X_t is the stochastic process counting the number in system at time t .

The time-homogenous process by excellence is the Markov Chain. We previously studied it in Chapter 2. In fact, this is the same definition of stationary distribution that we described in the study of inventory processes. Then, it is a reasonable goal to define a type of Markov Chains that can model manufacturing processes. The main class of Markov Chains used to model manufacturing processes are known as *Continuous Time Markov Chains* or CTMC. The simplest way to construct such chain requires two ingredients: First, a transition matrix P between states N , with elements p_{ij} denoting the probability of transitioning from state i to state j . Second, an average time the chain stays in a state that is also time homogeneous.

For a random time T to be time-homogeneous it is necessary that for any $s, t \geq 0$, we have $P(T \geq t) = P(T \geq t+s | T \geq s)$. Meaning that the tail CDF is the same regardless of time-shifts. Using Bayes's rule and rearranging yields the relation:

$$P(T \geq t)P(T \geq s) = P(T \geq t + s) \quad \text{or} \quad R(t)R(s) = R(t + s), \quad (7.2)$$

When defining $R(t) := P(T \geq t)$. Then, the goal is to find a distribution that satisfies this property. This implies that for $\Delta t = t/k$ we have $R(t) = R(\Delta t)^k$. In the discrete case, $P(T \leq i) = 1 - R(i + 1)$ and also $R(i + 1) = R(1)^{i+1}$. Putting these two relations together yields $P(T \leq i) = 1 - R(1)^{i+1}$, letting $1 - p = R(1)$, we will call this a *geometric distribution*.

In the continuous case, something very similar can be done. For example, $R(t) = R(1)^t$. Changing the base of the exponential¹ we have $R(t) = R(1)^t = e^{t \ln R(1)}$. Letting $\mu = -\ln(R(1)) > 0$ we have a distribution with tail cdf $R(t) = e^{-\mu t}$, we call this distribution the *exponential distribution*. See Figure 7.1 for a visual representation of the survival probabilities.

Having sorted out the time-homogeneous aspect of the average time in a state (what we need to do is simply calibrate the parameters of the discrete geometric distribution p or the exponential μ so that they match a target expected value). For example, in the exponential distribution case, we have that the average time in a state i , denoted as T_i is equal to $E(T_i) = \int_0^\infty e^{-\mu_i t} dt = \left[-\frac{1}{\mu_i} e^{-\mu_i t} \right]_0^\infty = 1/\mu_i$.

Now, we can think of a very small time interval Δt and think of the transition probability $P_{ij}(t)$ of our Markov Chain where time passes continuously. The first observation, is that from the memoryless property the probability of leaving a state is a constant times the time interval, for example, in the exponential case this is $\mu_i \Delta t$. Therefore, the probability of staying in the same state is $1 - \mu_i \Delta t$. If we extend this logic to all the states, we end up with this relation:

$$P_{ij}(t + \Delta t) \approx P_{ij}(t)(1 - \mu_j \Delta t) + \sum_{k \neq j} P_{ik}(t)(p_{kj} \mu_k \Delta t), \quad (7.3)$$

The interpretation of this equation is that at time $t + \Delta t$ one of two outcomes occurred: first, the chain simply stayed at state j . Or second, the chain transition to other state k , left that

¹ $x = e^{\log x}$, then $x^t = e^{t \ln x}$.

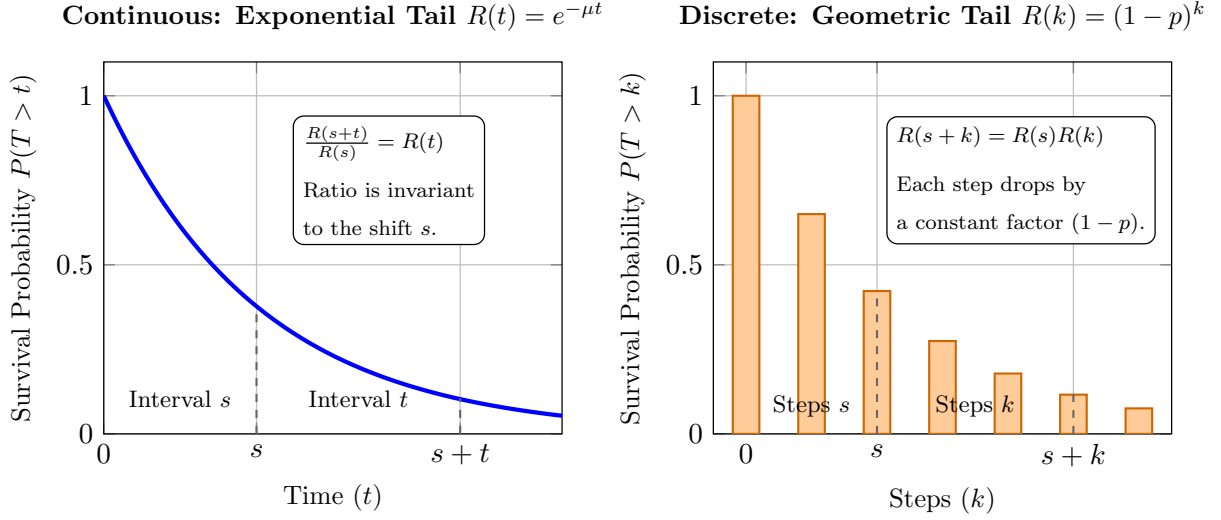


Figure 7.1: The visual architecture of memorylessness. The continuous Exponential curve (left) and discrete Geometric bars (right) represent the unique configurations where the relative remaining lifetime layout above a survival milestone s maintains a constant, self-similar proportion directly identical to the original starting profile at point 0.

state (with probability $\mu_k \Delta t$) to state j (with probability p_{kj}). Rearranging this term, we get:

$$\frac{P_{ij}(t + \Delta t) - P_{ij}(t)}{\Delta t} \approx -P_{ij}(t)\mu_j + \sum_{k \neq j} P_{ik}(t)(p_{kj}\mu_k), \quad (7.4)$$

Which is the definition of the derivative (or change) when Δt is small, leading to the equation:

$$\frac{dP_{ij}(t)}{dt} = -P_{ij}(t)\mu_j + \sum_{k \neq j} P_{ik}(t)(p_{kj}\mu_k), \quad (7.5)$$

When grouping all the elements, we get the equation $\frac{dP(t)}{dt} = P(t)Q$, where Q is a matrix with elements $q_{ii} = -\mu_i$ and $q_{ij} = p_{ij}\mu_i$. For readers familiar with calculus, the differential equation should be familiar: if we were to imagine $P(t)$ is a single variable equation (which it IS NOT, is a matrix), we could rearrange terms to $\frac{dP(t)}{P(t)} = Qdt$, integrate on both sides, getting $\ln P(t) - \ln P(0) = Qt$. Letting $P(0) = I$ (or actually the identity matrix), then $\ln(P(0)) = 0$. Taking exponential on both sides yields $P(t) = e^{Qt}$. Turns out that this intuition is also correct in the matrix case².

What we get from this is remarkable: the fact that to describe all the transitions we just need a matrix of constants Q , and that the transition is only proportional to the time interval Δt . The fact that $P(t) = e^{Qt}$ also gives us for free the equilibrium distribution. Assuming a very small time interval Δt , we have $P(\Delta t) \approx I + Q\Delta t$ from the Taylor expansion in the

²Instead of integrating, use the fact that $e^{Qt} = I + Qt + (Qt)^2/2! + (Qt)^3/3! + \dots$. Guess that $P(t) = e^{Qt}$, then $\frac{dP(t)}{dt} = Q + Q^2t + Q^3t^2/2! + \dots$. Factor the Q to get exactly $\frac{dP(t)}{dt} = QP(t)$. Lastly, $P(0) = I$, that is the system is in the initial state with probability 1.

previous footnote. We know that the equilibrium distribution follows:

$$\pi P(\Delta t) = \pi, \quad (7.6)$$

Then, we can just replace to get $\pi(I + Q\Delta t) = \pi$, which simplifies to the equilibrium distribution in CTMC:

$$\pi Q = \mathbf{0}, \quad (7.7)$$

Solving this system gives us exactly the equilibrium distribution of a system that evolves according to the probabilities p_{ij} and stays in state i for an exponential amount of time T_i .

As we studied in previous chapters, getting a close expression for the equilibrium distribution is a powerful modeling tool to understand the behaviour of a stochastic system. We illustrate this with an example:

Example 7.1. A strait (or a man-made waterway) is a passage connecting two large bodies of water. Imagine a country wants to inspect and charge ships as they cross. They can inspect up to 5 ships at the same time, in 5 stations. Each inspection takes in average $E(\tau) = 2$ hours per ship per station. The demand for the use of the strait depends on the fee ships are charged to pass. The random quantity of arriving ships a depends on this fee as $E(a) = e^{-5p}$ ships per hour where p is the fee a ship is charged per unit of time in the strait. Given p what is the average revenue that the country receives per hour assuming $a(p)$ ships arrive per hour? Also, what is the average amount of time a ship spends in this system? If instead of charging per hour, the country charges per ship, what's the new revenue?

The first step in solving this problem is defining the state space, that is, the set of states that we are going to analyze. From the statement of the problem there can be up to 5 ships in processing at any given time. Nonetheless, it is easy to imagine that a ship can arrive while 5 others are in processing, and potentially one more and so on. Also, the system can sometimes be empty. Then, it makes sense to let the state of the system be the integers $N = 0, 1, \dots$ representing the number of ships in the system. The strategy will be to estimate the average number of ships in the system (at any time), that is, $E(N) = 0\pi_0 + 1\pi_1 + 2\pi_2 \dots$. The system earns $pE(N)$ in the long term. To solve for π , we need to find it with $\pi Q = \mathbf{0}$. To do this, we simply need to describe how the system evolves: from any state with $i > 0$ ships in the system there are two possible outcomes, either a ship finish their service from a station or a new ship arrives. The arrival of ships is e^{-5p} per hour, so $\lambda_i := q_{i,i+1} = e^{-5p}$. Likewise, because of the memoryless property of the exponential distribution, the minimum of k exponential random variables $\min(\tau_1, \dots, \tau_k)$ is another exponential distribution³ with rate

³Note that $P(\min(\tau_1, \dots, \tau_k) \geq t) = P(\tau_1 \geq t, \dots, \tau_k \geq t) = P(\tau_1 \geq t) \dots P(\tau_k \geq t) = e^{-t/E\tau_1} \dots e^{-t/E\tau_k} =$

equal to the sum of the rates of the k exponentials. In our case, there are at most 5 stations or less active, then the transition occurs at a rate $\mu_i := q_{i,i-1} = \min(i, 5) \frac{1}{E(\tau)} = \min(i, 5)/2$. Finally, $q_{i,i} = -(q_{i,i-1} + q_{i,i+1})$. With this entries of Q solving the equilibrium $\pi Q = \mathbf{0}$ gives the equilibrium distribution of the system, from which $pE(N)$ is the rate the system earns.

This type of process is commonly called a *Birth-Death Process*, as from any state i , in principle there will be a transition to either $i + 1$ or $i - 1$ with rates λ_i and μ_i respectively. Computing the equilibrium distribution is also easy for these processes, as the equilibrium equation looks like:

$$\begin{aligned} -\pi_0\lambda_0 + \pi_1\mu_1 &= 0 \\ \pi_0\lambda_0 - \pi_1\mu_1 - \pi_1\lambda_1 + \pi_2\mu_2 &= 0 \\ &\vdots \end{aligned}$$

The first equation implies $\pi_0\lambda_0 = \pi_1\mu_1$, which in turn cancels out the same term in the second equation leaving $\pi_1\lambda_1 = \pi_2\mu_2$. The same continues occuring, thus having $\pi_k\lambda_k = \pi_{k+1}\mu_{k+1}$. Replacing $\pi_1 = \pi_0 \frac{\lambda_0}{\mu_1}$ and $\pi_2 = \pi_1 \frac{\lambda_1}{\mu_2} = \pi_0 \frac{\lambda_0\lambda_1}{\mu_1\mu_2}$ from which we can deduce that $\pi_k = \pi_0 \frac{\lambda_0\cdots\lambda_{k-1}}{\mu_1\cdots\mu_k}$. Lastly, since π must be a valid distribution (adding to 1) then we have $1 = \pi_0 + \pi_1 + \cdots$, then $1 = \pi_0[1 + \lambda_0/\mu_1 + \cdots]$. Which yields $\pi_0 = [1 + \lambda_0/\mu_1 + \cdots]^{-1}$. After calculating these, the average money that the country earns is $pE(N) = p[0\pi_0 + 1\pi_1 + 2\pi_2 + \cdots]$.

Estimating the average waiting time seems hard, as in principle it would require to average many possible paths for different number of ships in the system. But we can try to use what we know to make this estimation. For any ship arriving at random, this ship will experience a random amount of time in the system W . This ship would have to pay $pE(W)$ in average. In an hour, there are $E(a) = e^{-5p}$ ships arriving per hour, then, the amount they pay should be $E(q)pE(W)$. Note that the amount the ships pay is in fact the same revenue we found before. Then, it must be true that $pE(N) = E(q)pEW$. This is known as *Little's law*:

$$E(W) = \frac{E(N)}{E(a)}, \quad (7.8)$$

This law is extremely useful, because we got the expected value of the waiting time $E(W)$ for free, just knowing the arrival rate $E(a)$ and the average number in the system $E(N)$. In general, with this relation, we need only two of the quantities to guess the third. In most textbooks this law is stated as $E(N) = E(a)E(W)$. In Figure 7.2 there is a graphical representation of the principle.

$e^{-t[1/E\tau_1 + \cdots + 1/E\tau_k]}$. That is, an exponential distribution with a rate equal to the sum of the rates of the other exponentials.

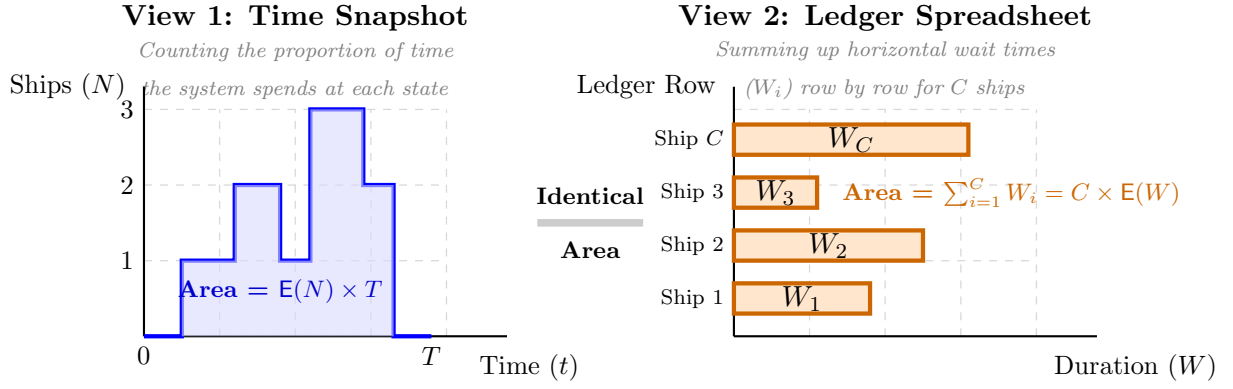


Figure 7.2: Geometric derivation of Little's Law by equating total system revenue over an observation interval T . View 1 scales the time-weighted average number of ships in the system ($E(N) \times T$) by a fee p . View 2 scales the spreadsheet accounting record of individual random turnaround times experienced by C arriving ships ($\sum W_i = C \times E(W)$) by the same fee p . Equating both operational perspectives balances the ledger, yielding $pE(N)T = pCE(W)$, which simplifies directly to $E(N) = E(a)E(W)$ given the average arrival rate $E(a) = C/T$.

For the last situation, the country notes an interesting fact: as long as the system is flowing, that is, the rate of arrivals $E(a)$ is lower than the maximum rate of service $5/E(\tau)$. Meaning that the line of ships will not grow indefinitely, it must be the case that the rate of departures is the same as the rate of arrivals. The logic follows from the fact that it cannot be the case that the rate of departures is higher than the rate of arrivals as this would imply that somehow more ships are created during the inspection, which is impossible. On the other hand, the rate of departures cannot be less than that of departures as it would imply a collapse of the system longterm (remember, everything is time-homogenous so far).

If we think about why this is the case, recall that in the derivation of the equilibrium distribution, we had that for every state $\pi_k \lambda_k = \pi_{k+1} \mu_{k+1}$. Which in plain words means that in equilibrium between two states, the inflow and outflow of that partition is exactly the same. This property is often called *detailed balance*⁴ and means that the chain behaves the same forward or backwards (think that there is a perfect symmetry of the flows forwards or backwards). Imagining an alternative process, where time is reversed, that is arrivals are departures and departures are arrivals, will distribute exactly the same. The key is that there is a strong symmetry that allows this.

The conclusion is extremely powerful, because if departures behave the same probabilistically than arrivals then it means that when you connect queues together, you can study them each

⁴This property is described as $\pi_k q_{k,k+1} = \pi_{k+1} q_{k+1,k}$

individually as the output will behave the same as the input (in this case a so-called Poisson process, which only means that the time between arrivals follows an exponential distribution).

If for example the country wished to connect multiple queues in tandem (say ℓ) or in a network where the next tandem is selected with a fixed probability (a so-called routing matrix), starting with Poisson arrivals at a rate $\mathbf{E}(a) = e^{-5p}$. Then, the equilibrium distribution of the number in each queue is just the product of the equilibrium distributions of each queue as:

$$\pi(N_1, \dots, N_\ell) = \pi_1(N_1) \times \dots \times \pi_\ell(N_\ell), \quad (7.9)$$

Likewise, this also implies that after the tandem of queues, the departure rate is identical in distribution to the arrivals. Therefore, the revenue of charging ships only once is $p\mathbf{E}(a) = pe^{-5p}$. The price that maximizes the revenue is given by equating the derivative to 0, that is, $pe^{-5p}5 = e^{-5p}$ or $p = 1/5$.

7.3 Time-varying arrivals

Time homogeneity (in both arrivals and processing times) is often an unrealistic assumption. Many manufacturing processes are subject to time-varying demand patterns. For example, ports have different demand patterns depending of the time of the year and time of day. Likewise, certain manufacturing processes see a spike of the demand at certain times of the year. Toy manufacturers see an increase towards the holidays. Certain rare events such as pandemics increase the demand for medical supplies and so on. All of these examples emphasize the need to extend the analysis to systems where the arrival changes over time.

The need to consider time-varying arrivals seem to contradict the time-homogenous structure of the (Continuous) Markov Chains that we have studied. To see why, suppose that the random arrivals a change in average over time. From our equilibrium analysis, we get away by considering time by having that the interarrival times follow an exponential distribution, where the probability of seeing a new arrival in an interval Δt is $\lambda\Delta t$ for an exponential random variable with rate μ . If we wanted to change this rate to be time dependent as $\lambda(t)$ it would break the equilibrium conditions everywhere, as $\frac{dP(t)}{dt}$ is not a constant anymore.

Before throwing the time-homogenous time into the garbage, one can imagine a way to salvage the model, while still allowing some degree of change in the arrivals: The strategy is to consider dividing time into relatively *long* intervals where the arrivals remain with a constant rate r_0 and then change to another rate r_1 . For example, this can happen every 12 hours. You can extend the strait example to imagine that there is a higher demand rate r_0 during the daytime hours (say 7am-7pm) than during the other 12 night hours.

We can imagine in this mental experiment, the system is going to behave almost like the

homogeneous case in the middle of each period. For example, at noon the system pretty much should look as if the arrival rate is exactly r_1 all the time. The problem is going to be at the times that the rate changes. But if we are willing to ignore this, then the approximation should still work with some caveats. For example, Little's Law states that the average number in system $E(N)$ is equal to $E(W)E(a)$. In this case, we should expect that $E(N) = E(W)r_0$ during the daytime range and $E(N) = E(W)r_1$ during the night, assuming the service rate is also the same. If the processing is very fast compared to the arrival rate, meaning that the system is empty most of the time. Then, these approximations should hold relatively well. The problem is that if the processing is slow, if $r_0 > r_1$, at the beginning of the night shift at 7pm there might be still jobs in process from the earlier part of the day.

This phenomenon is in fact very common in a lot of manufacturing and retail systems: processing is high-traffic, that is the system sees a number of arrivals that it can barely process. In a way, this makes economic sense, as a faster system means spending more money on processing resources (workers, servers, stations, etc), and unless there is a reward for higher throughput, the company will spend the minimum amount of resources that allows to meet the demand or target throughput. In other words, it makes economic sense to have a system that is barely stable. In a way it makes sense that some unprocessed work accumulates over less busy periods.

Going back to our mental experiment, at 7am the system should start almost empty. At 7pm, letting $t = 0$, suppose that the system has $N(0) > 0$ jobs that are still in processing when the arrival rates changes to the lower intensity r_1 . Then, we would like to know how many jobs are in the system at 8pm, that is, $t = 1$. We can do multiple analyses. First we start with the case where there are infinite servers (while unrealistic, there are some systems that behave like this). Suppose, each server servers at a rate μ . See Figure 7.3 for a visual representation.

7.3.1 Time-varying arrivals with infinite servers

Since there are always available servers, we can treat the two populations separately: on one hand the previous jobs $N(0)$ plus the new jobs that arrive between 7pm and 8pm, that is, from $t = 0$ to $t = 1$. For the new jobs, because of the memoryless property, their time to service is the exponential rate. At $t = 1$ we have:

$$E \left[\sum_{i=1}^{N(0)} \mathbf{1}\{T_i \geq t = 1\} \right] = \sum_{i=1}^{N(0)} P(T_i \geq 1) = N(0)e^{-1\mu}, \quad (7.10)$$

For the new arrivals, between $t = 0$ and $t = 1$ we have that at each small interval dt there is a new arrival with rate r_1 and each of these arrivals will still be in the system at 8 pm, that is, $t = 1$ with probability equal to the exponential. For example, an arrival at $s \in (0, 1)$ will still be in processing with probability $e^{-(1-s)\mu}$, which is exactly the definition of the reliability function $R(1 - s)$. Since at each dt there is an arrival with probability $r_1 dt$, and the interval is

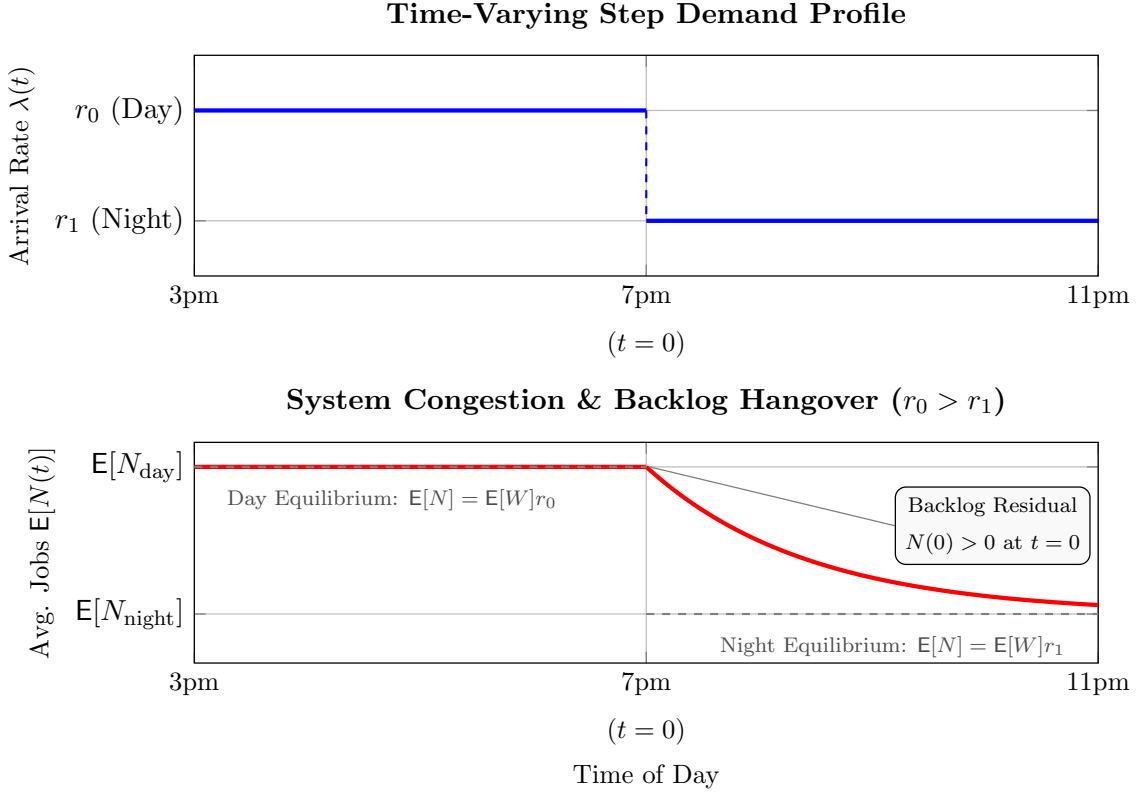


Figure 7.3: The mechanics of a non-homogeneous queueing system switching operational regimes. The upper axis illustrates a piece-wise constant arrival pattern splitting daytime demand (r_0) from nighttime demand (r_1). The lower axis visualizes the transient response of the expected inventory $E[N(t)]$. If processing speeds are tight, the system creates an inventory carryover at the 7pm boundary ($t = 0$), meaning remnants of daytime congestion bleed into the night shift.

so small that the probability that 2 arrivals occur is 0. We can do this calculation for each dt as:

$$\int_0^1 e^{-(1-s)\mu} r_1 ds = r_1 \int_0^1 e^{-(1-s)\mu} ds = r_1 \int_0^1 e^{-s\mu} ds = \frac{r_1}{\mu} (1 - e^{-\mu}), \quad (7.11)$$

Now that we know the expected numbers from each population (the new and old), we can put them together as:

$$E[N(1)|N(0)] = N(0)e^{-1/E(W)} + E(a(0,1))E(W)(1 - e^{-1/E(W)}) \quad (7.12)$$

Where $E(W) = \mu^{-1}$ and $E(a(0,1)) = r_1$. This equation can be seen as a transient version of *Little's Law* where when $E(W) \rightarrow 0$, meaning that the system is fast enough to process the jobs quickly and the system just converges to the expected Little's Law (as the term $e^{-1/E(W)}$ vanishes) and all that is left is $E[N(1)|N(0)] = E(a(0,1))E(W)$. Since this relation applies between any two time intervals, we can apply it to systems when the arrival rates change (and remain constant) in known intervals.

Then, for a situation when the rates change every period, say $r_1 = E(a(0,1))$, $r_2 = E(a(1,2)), \dots$, we can conclude the following relation:

$$E[N(t)|N(t-1)] = N(t-1)e^{-1/E(W)} + r_t E(W)(1 - e^{-1/E(W)}), \quad (7.13)$$

If we take expectation on both sides, we end up with the relation:

$$E[N(t)] = E[N(t-1)]\alpha + \beta r_t, \quad (7.14)$$

With $\alpha = e^{-1/E(W)}$ and $\beta = E(W)(1 - \alpha)$. Solving this equation recursively starting at $N(0)$, so that $E[N(1)] = \alpha N(0) + \beta r_1$, $E[N(2)] = \alpha(\alpha N(0) + \beta r_1) + \beta r_2$ and so on, yields $E[N(t)] = \alpha^t N(0) + \beta \sum_{i=1}^t \alpha^{i-t} r_i$, or the relation:

$$E[N(t)] = e^{-t/E(W)} N(0) + E(W)(1 - e^{-1/E(W)}) \sum_{i=1}^t r_i e^{-(i-t)/E(W)}, \quad (7.15)$$

Intuitively this result can be seen as an average over the entire history of the process. The first term represents the oldest arrivals starting at time 0, and each term of the summation is the contribution of arrivals each time step. Similar as before, when $E(W) \rightarrow 0$, all the terms of the summation vanish, except the last one (as $e^{-(t-t)/E(W)} = 1$), leaving $E[N(t)] = E(W)r_t$ which is Little's Law as expected.

This kind of recursion is useful when we have a system that has stable service rate, but time varying arrivals. These arrivals can be a seasonal demand that can be estimated with the methods in Chapter 1 and we want to know how much resources are needed to sustain the operation in the future: imagine a factory with fixed and enough machinery which staffing levels are proportional to the number of jobs in the system $E[N(t)]$ for a distant t . This recursion would allow us to estimate the staffing load knowing the number of jobs today at $N(0)$.

7.3.2 Time-varying arrivals with finite servers

The infinite server case is a useful approximation in cases where the servers are not the bottleneck of the system, that is, relative to the amount of arrivals there is always enough processing capacity. This is the case in certain manufacturing and computing systems. When the processing capacity is limited by the number of servers, say k of them each serving at a rate $\mu = 1/E(W)$, the situation is different. Starting the $N(0) > 0$ jobs in queue, when $N(0) \leq k$ we are in a similar situation to the infinite server case, as the number in system will keep decreasing at an exponential rate towards equilibrium to Little's law.

Suppose the arrival rate $E(a(0,1))$ is again r_1 and $N(0) > k$ meaning that the system starts with more jobs than available servers. At maximum capacity the effective service rate is $k\mu$ (all servers are active). In this case, the system will behave linearly, with arrivals growing at a rate of r_1 and exiting at a rate $k\mu$. In this case, the case behaves in a linear fashion as the number in system is $(r_1 - k\mu)t$. If $r_1 > k\mu$ the queue will keep growing as the server at maximum

capacity is not able to process jobs fast enough to vacate the queue. Likewise, when $r_1 \leq k\mu$, the number in system will decrease linearly until either the arrival rate changes at $t = 1$ to r_2 or the number in the system is equal to k . More explicitly, for a time $t \in (0, 1)$ the number in system is in average $N(t) = N(0) + (r_1 - k\mu)t$.

If for example, $(r_1 - k\mu)t$ is decreasing fast enough it could happen that there is a $t < 1$ that the system now has less than k jobs, that is, a $N(t) < k$. In this case, as jobs get processed the system becomes exactly like the infinite server case as jobs that arrive will find a server and will behave statistically the same way as the previous section. This time will be $t^* = \frac{k-N(0)}{r_1-k\mu}$ if this is less than 1, then the change in the dynamic will occur by then and the system will start behaving as the infinite server regime. That is, $E[N(1)] = E[N(t^*)]e^{-(1-t^*)/E(W)} + r_1E(W)(1 - e^{-(1-t^*)/E(W)})$. This can be seen as restarting the system at time t^* .

As the rates change the system can alternate between the linear regime $E[N(t)] > k$ or the exponential regime $E[N(t)] \leq k$. Even though it is called exponential regime, it does not mean that it is faster than the linear one. In fact, it is slower as less than k servers are in operation while the arrival rate remains the same. For a system starting at time $t - 1$ with $N(t - 1)$ jobs there are the following cases:

- $N(t - 1) > k$ a linear system: The system evolves as $E[N(t)] = E[N(t - 1)] + (r_t - k\mu)t$ either until the next rate change or until a time $\tau > t - 1$ when the system enters the exponential regime.
- $N(t - 1) \leq k$ an exponential system: The system evolves as $E[N(t)] = E[N(t - 1)]\alpha(\Delta t) + (1 - \alpha(\Delta t))E(W)r_t$ until the change of arrival rate or until a time $\tau > t - 1$ when the system enters the linear regime.

We illustrate this with an example:

Example 7.2 (World Cup Tickets). A cloud provider service hosts an application to sell World Cup tickets. They have staggered access to the website starting at 9am to 10 thousand premium customers, the system starts with $N(0) = 5$ thousand and the arrival rate for the next hour is the other 5 thousand. The general population will have access starting at 10am, arriving at a rate of 20 thousand per hour until noon when the sale closes. Each person takes in average 10 minutes to complete their purchase. The cloud computing company wants to know how much computing power to allocate. They can choose between two configurations $k = 3$ or $k = 5$ thousand. Plot the average number in the system.

The first aspect to determine is in what regime the system starts. In the case of the configuration with $k = 5$ thousand servers the system starts in the exponential system. As time progresses, the average number in system $E[N(t)]$ converges to Little's law: The average

time in system is $E[W] = 1/\mu = 1/6$ and the arrival rate is $E(a(0,1)) = 5$ thousand. Then, the system decays until convergence at $E(W)E(a(0,1)) = 5/6$ thousand (approximately 833 customers). This will continue until time $t = 1$, corresponding to 10am when the arrival rate $E(a(1,3))$ changes to 20 thousand per hour. In this case, since we start at the exponential regime at time 1 with 833 average customers. The new equilibrium implied by Little's law is $E(a(1,3))E(W) = 20/6$ thousand (approximately 3,333 customers). Since this is below $k = 5000$, the system should continue in the exponential regime until closing at time 3 (noon).

The other configuration is slightly more interesting. First, the system starts in the linear regime as $N(0) = 5$ thousand is greater than the available $k = 3$ thousand servers. The slope of the system is $E(a(0,1)) - k\mu = 5 - 3(6) = -13$ thousand customers per hour. Starting at $N(0) = 5$ the time that the system reach the exponential regime is $5 - 13t = 3$ or $t = 2/13 \approx 0.15$ or around 9:10am. At this time the system enters the exponential regime and converges again to the equilibrium point of 833 customers. Likewise, at 10am the system starts in the exponential regime towards 3,333 customers. But before that, it must reach 3000 (the number of servers) at a time $e^{-6t}(5/6) + (1 - e^{-6t})(20/6) = 3$ or just $2/6 = e^{-6t}(15/6)$ which simplifies to $t = \ln(15/2)/6 \approx 0.33$ or around 10:20am. At this time, the system enters the linear regime growing a rate of $E(a(1,3)) - k\mu = 20 - 3(6) = 2$ thousand customers per hour. This linear growth will continue until the sales closing time at noon, the system will never recover or reach equilibrium. See Figure 7.4.

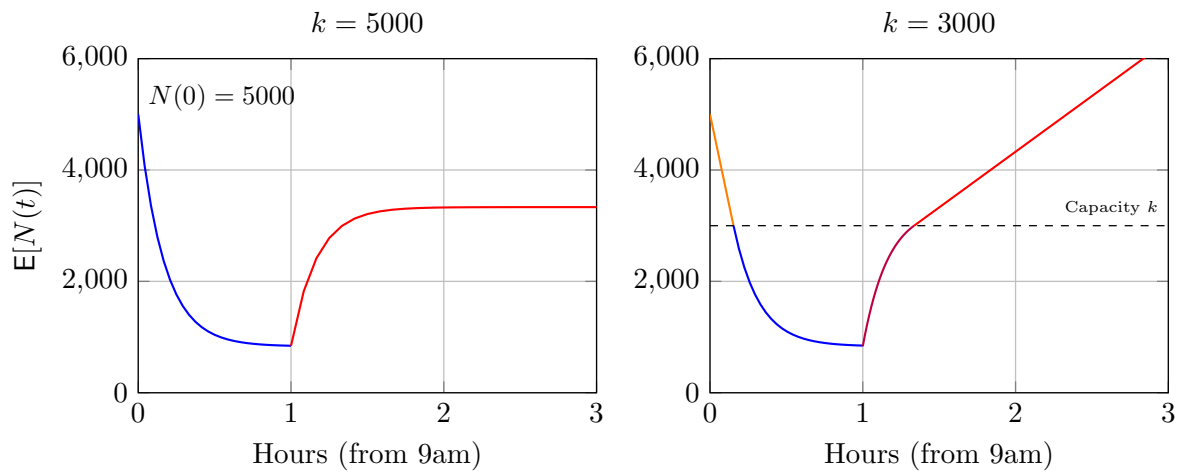


Figure 7.4: Transient behavior of the average number of customers in the system $E[N(t)]$ over a 3-hour window for two capacity configurations. The left plot ($k = 5000$) shows a system that remains entirely in the exponential regime, successfully absorbing both the initial workload and the 10am demand surge well below capacity limits. The right plot ($k = 3000$) exhibits alternating regimes: an initial linear drain down to the capacity boundary, a stable exponential phase chasing an 833-customer equilibrium, followed by a post-10am surge that crosses back over the capacity line into an unsustainable, linear growth phase.